

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VIII

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Sorting Algorithms:

Write programs to implement and compare the following sorting algorithms:

a. Bubble sort

CODE:

```
def bubbleSort(array):  
  
    for i in range(len(array)):  
  
        for j in range(0, len(array) - i - 1):  
  
            if array[j] > array[j + 1]:  
  
                temp = array[j]  
                array[j] = array[j+1]  
                array[j+1] = temp  
  
data = [-2, 45, 0, 11, -9]  
  
bubbleSort(data)  
  
print('Sorted Array in Ascending Order:')  
print(data)  
  
print("Aniket")
```

```

def bubbleSort(array):
    for i in range(len(array)):
        swapped = False
        for j in range(0, len(array) - i - 1):
            if array[j] < array[j + 1]:
                temp = array[j]
                array[j] = array[j+1]
                array[j+1] = temp
                swapped = True
        if not swapped:
            break

data = [-2, 45, 0, 11, -9]

bubbleSort(data)

print('Sorted Array in Descending Order:')
print(data)

print("Aniket")

```

OUTPUT:

```

===== RESTART:
Sorted Array in Ascending Order:
[-9, -2, 0, 11, 45]
Aniket
===== RESTART:
Sorted Array in Descending Order:
[45, 11, 0, -2, -9]
Aniket

```

b. Insertion sort

CODE:

```
def insertionSort(array):  
  
    for step in range(1, len(array)):  
        key = array[step]  
        j = step - 1  
  
        while j <= 0 and key < array[j]:  
            array[j + 1] = array[j]  
            j = j - 1  
  
        array[j + 1] = key  
  
data = [9, 5, 1, 4, 3]  
insertionSort(data)  
print('Sorted Array in Ascending Order:')  
print(data)  
print("Aniket")
```

```
def insertionSort(array):  
  
    for step in range(1, len(array)):  
        key = array[step]  
        j = step - 1  
  
        while j >= 0 and key > array[j]:  
            array[j + 1] = array[j]  
            j = j - 1  
  
        array[j + 1] = key  
  
data = [9, 5, 1, 4, 3]  
insertionSort(data)  
print('Sorted Array in Descending Order:')  
print(data)  
print("Aniket")
```

OUTPUT:

```
===== RESTART: C:/Users
Sorted Array in Ascending Order:
[1, 3, 4, 5, 9]
Aniket
```

```
===== RESTART: C:/Users/
Sorted Array in Descending Order:
[9, 5, 4, 3, 1]
Aniket
```

c. Selection sort

CODE:

```
def selectionSort(array, size):

    for step in range(size):
        min_idx = step

        for i in range(step + 1, size):

            if array[i] < array[min_idx]:
                min_idx = i

        (array[step], array[min_idx]) = (array[min_idx], array[step])

data = [-2, 45, 0, 11, -9]
size = len(data)
selectionSort(data, size)
print('Sorted Array in Ascending Order:')
print(data)

print("Aniket")
```

```
def selectionSort(array, size):

    for step in range(size):
        min_idx = step

        for i in range(step + 1, size):

            if array[i] > array[min_idx]:
                min_idx = i

        (array[step], array[min_idx]) = (array[min_idx], array[step])

data = [-2, 45, 0, 11, -9]
size = len(data)
selectionSort(data, size)
print('Sorted Array in Descending Order:')
print(data)

print("Aniket")
```

OUTPUT:

```
----- RESTART:
Sorted Array in Ascending Order:
[-9, -2, 0, 11, 45]
Aniket
```

```
===== RESTART:
Sorted Array in Descending Order:
[45, 11, 0, -2, -9]
Aniket
```